

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12383640
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Brockschmidt, Jr.; Arthur E. et al.

Ultraviolet light-emitting assembly

Abstract

Assemblies and methods for disinfecting surfaces using ultraviolet (UV) light are disclosed. In one aspect, a UV light-emitting assembly comprises a plurality of UV light emitters, and first and second UV light emitter supports seating the UV light emitters. A first heat sink is affixed to the first UV light emitter support and a second heat sink is affixed to the second UV light emitter support. A thermally conductive and electrically insulating plate contacts the first heat sink and the second heat sink.

Inventors: Brockschmidt, Jr.; Arthur E. (Renton, WA), Childress; Jamie J. (Mercer Island, WA)

Applicant: The Boeing Company (Chicago, IL)

Family ID: 78806331

Assignee: The Boeing Company (Arlington, VA)

Appl. No.: 17/452564

Filed: October 27, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220184251 A1	Jun. 16, 2022

Related U.S. Application Data

us-provisional-application US 63154239 20210226
us-provisional-application US 63124341 20201211

Publication Classification

Int. Cl.: A61L2/10 (20060101); **A61L2/26** (20060101); **F21V21/14** (20060101); **F21V23/00** (20150101); **F21V29/74** (20150101)

U.S. Cl.:

CPC A61L2/10 (20130101); **A61L2/26** (20130101); **F21V21/14** (20130101); **F21V23/003** (20130101); **F21V29/74** (20150115); **A61L2202/11** (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5498394	12/1995	Matschke	N/A	N/A
5594304	12/1996	Graber	N/A	N/A
5660719	12/1996	Kurtz et al.	N/A	N/A
6144175	12/1999	Parra	N/A	N/A
6656424	12/2002	Deal	N/A	N/A
7595723	12/2008	Heitzmann et al.	N/A	N/A
8581522	12/2012	Inskeep	N/A	N/A
8791441	12/2013	Lichtblau	N/A	N/A
8834805	12/2013	Owen et al.	N/A	N/A
9623133	12/2016	Childress et al.	N/A	N/A
11382993	12/2021	Childress	N/A	N/A
2004/0249369	12/2003	Muzzi et al.	N/A	N/A
2005/0152146	12/2004	Owen	362/294	F21V 29/57
2005/0158206	12/2004	Moisan	422/186.21	A61L 2/14
2008/0152548	12/2007	Clark et al.	N/A	N/A
2008/0260601	12/2007	Lyon	N/A	N/A
2008/0289800	12/2007	Simadiris et al.	N/A	N/A
2009/0045750	12/2008	Brggs et al.	N/A	N/A
2011/0243789	12/2010	Roberts	N/A	N/A
2012/0051030	12/2011	Johnson	N/A	N/A
2012/0076702	12/2011	Dunkley et al.	N/A	N/A
2012/0168641	12/2011	Lizotte	N/A	N/A
2012/0240968	12/2011	Schumacher	N/A	N/A
2012/0305787	12/2011	Henson	N/A	N/A
2013/0330235	12/2012	Stibich et al.	N/A	N/A
2014/0044590	12/2013	Trapani	N/A	N/A
2014/0175280	12/2013	Tantillo	N/A	N/A
2014/0320009	12/2013	Goscha et al.	N/A	N/A
2015/0064065	12/2014	Kreitenberg	N/A	N/A
2015/0165079	12/2014	Shur et al.	N/A	N/A
2015/0182647	12/2014	Ranta et al.	N/A	N/A
2015/0235727	12/2014	Lott et al.	N/A	N/A

2016/0271288	12/2015	Davis	N/A	N/A
2016/0324996	12/2015	Bilenko et al.	N/A	N/A
2017/0198896	12/2016	May	N/A	N/A
2018/0064833	12/2017	Childress et al.	N/A	N/A
2018/0133351	12/2017	Smetona et al.	N/A	N/A
2018/0243582	12/2017	Kaneda et al.	N/A	N/A
2018/0339075	12/2017	Kennedy et al.	N/A	N/A
2019/0142981	12/2018	Kim et al.	N/A	N/A
2019/0300174	12/2018	Young et al.	N/A	N/A
2020/0061223	12/2019	Hallack	N/A	N/A
2020/0234941	12/2019	Yagyu et al.	N/A	N/A
2020/0289698	12/2019	Polidoro	N/A	N/A
2021/0213147	12/2020	Donhowe et al.	N/A	N/A
2021/0338860	12/2020	Grenon et al.	N/A	N/A
2021/0346540	12/2020	Childress et al.	N/A	N/A
2021/0346541	12/2020	Callahan et al.	N/A	N/A
2021/0346561	12/2020	Callahan et al.	N/A	N/A
2021/0361794	12/2020	Yencho	N/A	N/A
2021/0386882	12/2020	Brockschmidt, Jr. et al.	N/A	N/A
2021/0386883	12/2020	Childress	N/A	N/A
2021/0386884	12/2020	Brockschmidt, Jr. et al.	N/A	N/A
2021/0396918	12/2020	Gross et al.	N/A	N/A
2022/0023458	12/2021	Brockschmidt, Jr. et al.	N/A	N/A
2022/0023459	12/2021	Colletti et al.	N/A	N/A
2022/0023478	12/2021	Childress	N/A	N/A
2022/0068627	12/2021	Yagyu et al.	N/A	N/A
2022/0111086	12/2021	Childress	N/A	N/A
2022/0111087	12/2021	Childress et al.	N/A	N/A
2022/0111096	12/2021	Childress	N/A	N/A
2022/0113006	12/2021	Childress et al.	N/A	N/A
2022/0133925	12/2021	Gray et al.	N/A	N/A
2022/0184252	12/2021	Childress et al.	N/A	N/A
2023/0258315	12/2022	Shinoda	252/588	H01J 61/30

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2967190	12/2015	CA	N/A
3123597	12/2019	CA	N/A
202342524	12/2011	CN	N/A
106049005	12/2015	CN	N/A
211204004	12/2019	CN	N/A
111744026	12/2019	CN	N/A
3910653	12/1989	DE	N/A
H09201401	12/1996	JP	N/A
2000283840	12/1999	JP	N/A

2002100323	12/2001	JP	N/A
2018055965	12/2017	JP	N/A
2018092755	12/2017	JP	N/A
2018167166	12/2017	JP	N/A
1020120083541	12/2011	KR	N/A
20150045628	12/2014	KR	N/A
2012142427	12/2011	WO	N/A
2015028334	12/2014	WO	N/A
2016210399	12/2015	WO	N/A
2020052506	12/2019	WO	N/A

OTHER PUBLICATIONS

Souza et al., Study of the Influence of Variation in Distances Between Electrodes in Spectral DBD Plasma Excitation, Materials Research. 2016; 19(1): 202-206 (Year: 2016). cited by examiner

United States Patent and Trademark Office, Office Action Issued in U.S. Appl. No. 17/452,559, filed Feb. 1, 2024, 39 pages. cited by applicant

European Patent Office, Office Action Issued in Application No. 21210726.2, Sep. 14, 2023, Germany, 9 pages. cited by applicant

European Patent Office, Office Action Issued in Application No. 21210723.9, Sep. 19, 2023, Germany, 5 pages. cited by applicant

European Patent Office, Office Action Issued in Application No. 21210728.8, Sep. 20, 2023, Germany, 5 pages. cited by applicant

Plass, C. et al., "Ultraviolet Wand," U.S. Appl. No. 29/735,235, filed May 19, 2020, 15 pages. cited by applicant

Barrett, L. et al., "Sterilization of sea lice eggs with ultraviolet C light: towards a new preventative technique for aquaculture," Post Management Science, vol. 76, No. 3, Mar. 2020, 7 pages. cited by applicant

European Patent Office, Extended European Search Report issued in application No. 21202296.6, May 24, 2022, Germany, 11 pages. cited by applicant

European Patent Office, Extended European Search Report issued in application No. 21207466.0, May 2, 2022, Germany, 8 pages. cited by applicant

European Patent Office, Extended European Search Report issued in application No. 21210410.03, Apr. 25, 2022, Germany, 10 pages. cited by applicant

Honeywell UV Treatment System, Available online at <https://aerospace.honeywell.com/en/learn/products/cabin/uv-cabin-system>, Available as early as Jun. 21, 2021, 4 pages. cited by applicant

Huang, B. et al., "Research on UV radiation measurements and correction methods," Proceedings of the International Symposium on Photoelectronic Detection and Imaging, May 24, 2011, Beijing, China, 10 pages. cited by applicant

ISA United States Patent Office, Written Opinion issued in application No. PCT/US2016/039506, May 31, 2017, WIPO, 8 pages. cited by applicant

European Patent Office, partial European Search Report issued in application No. 21202296.6, Mar. 4, 2022, Germany, 11 pages. cited by applicant

"UVC Sensors to Monitor Ultraviolet Germicidal Irradiation {UVGI}," Available online at <https://www.pro-lite.co.uk/File/Solar%20Light%20UVGI%20Radiometer%20Brochure.pdf>, Jan. 2019, 2 pages. cited by applicant

European Patent Office, Partial European Search Report Issued in Application No. 21210723.9, May 12, 2022, Germany, 16 pages. cited by applicant

United States Patent and Trademark Office, Office Action Issued in U.S. Appl. No. 17/452,561, filed Jan. 3, 2024, 60 pages. cited by applicant

European Patent Office, Extended European Search Report Issued in Application No. 21210725.4, Sep. 6, 2022, Germany, 79 pages. cited by applicant
European Patent Office, Extended Search Report Issued in Application No. 21210726.2, Sep. 8, 2022, Germany, 52 pages. cited by applicant
European Patent Office, Extended Search Report Issued in Application No. 21210727.0, Sep. 9, 2022, Germany, 16 pages. cited by applicant
European Patent Office, Extended Search Report Issued in Application No. 21210723.9, Sep. 23, 2022, Germany, 21 pages. cited by applicant
European Patent Office, Extended European Search Report Issued in Application No. 21210728.8, Sep. 28, 2022, Germany, 13 pages. cited by applicant
Welch, D. et al., "Far-UVC light: A new tool to control the spread of airborne-mediated microbial diseases," Scientific Reports, vol. 8, No. 1, Feb. 9, 2018, 7 pages. cited by applicant
"Ushio Care222, Far UV-C Disinfection Technology," Ushio, Oct. 3, 2020, 5 pages. cited by applicant
European Patent Office, Partial European Search Report Issued in Application No. 21210725.4, Apr. 26, 2022, Germany, 16 pages. cited by applicant
European Patent Office, Partial European Search Report Issued in Application No. 21210726.2, May 3, 2022, Germany, 17 pages. cited by applicant
European Patent Office, Partial European Search Report Issued in Application No. 21210727.0, May 4, 2022, Germany, 16 pages. cited by applicant
Buonanno, M. et al., "Far-UVC light (222nm) efficiently and safely inactivates airborne human coronaviruses," Scientific Reports, vol. 10, Jun. 24, 2020, 8 pages. cited by applicant
European Patent Office, Partial European Search Report Issued in Application No. 21210728.8, Jun. 9, 2022, Germany, 14 pages. cited by applicant
United States Patent and Trademark Office, Notice of Allowance Issued in U.S. Appl. No. 17/452,560, filed Mar. 1, 2024, 41 pages. cited by applicant
United States Patent and Trademark Office, Notice of Allowance Issued in U.S. Appl. No. 17/452,561, filed Apr. 2, 2024, 19 pages. cited by applicant
United States Patent and Trademark Office, Office Action Issued in U.S. Appl. No. 17/452,555, filed Nov. 22, 2024, 53 pages. cited by applicant
United States Patent and Trademark Office, Office Action Issued in U.S. Appl. No. 17/464,792, filed Jan. 16, 2025, 46 pages. cited by applicant

Primary Examiner: Kipouros; Holly

Attorney, Agent or Firm: Alleman Hall & Tuttle LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/154,239, filed Feb. 26, 2021, and to U.S. Provisional Patent Application Ser. No. 63/124,341, filed Dec. 11, 2020, the entirety of which are hereby incorporated herein by reference for all purposes.

FIELD

(1) This disclosure generally relates to disinfecting surfaces, and more particularly to assemblies and methods for disinfecting surfaces using ultraviolet (UV) light.

BACKGROUND

(2) Ultraviolet (UV) light has been used in some settings to disinfect and sanitize surfaces. In some

examples, multiple UV emitters are provided and powered by a relatively low power supply, such as 12 watts. While such UV devices offer promise in their ability to render inactive and/or kill certain pathogens, challenges exist in developing devices and systems for more effective delivery of such UV radiation.

SUMMARY

(3) According to one aspect, an ultraviolet (UV) light-emitting assembly is provided that comprises a plurality of UV light emitters, a first UV light emitter support seating the plurality of UV light emitters, and a second UV light emitter support seating the plurality of UV light emitters. The first UV light emitter support is spaced by a gap from the second UV light emitter support. A first heat sink is affixed to the first UV light emitter support and a second heat sink affixed to the second UV light emitter support. A thermally conductive and electrically insulating plate contacts the first heat sink and the second heat sink.

(4) According to another aspect, an ultraviolet (UV) light-emitting assembly is provided that comprises a plurality of UV light emitters, a first UV light emitter support seating the plurality of UV light emitters, and a second UV light emitter support seating the plurality of UV light emitters. The first UV light emitter support is spaced by a gap from the second UV light emitter support. A moving mechanism is configured to vary the gap between the first UV light emitter support and the second UV light emitter support to thereby change an intensity of UV light emitted from the plurality of UV light emitters.

(5) According to another aspect, an ultraviolet (UV) light-emitting assembly is provided that comprises a plurality of UV light emitters, a first UV light emitter support seating the plurality of UV light emitters, and a second UV light emitter support seating the plurality of UV light emitters. The first UV light emitter support is spaced from the second UV light emitter support. The assembly further includes a heat sink comprising an actively-cooled plate. A first thermally conductive and electrically insulating pad comprises a first upper face that contacts a first bottom face of the first UV light emitter support and a first lower face that contacts the actively-cooled plate. A second thermally conductive and electrically insulating pad comprises a second upper face that contacts a second bottom face of the second UV light emitter support and a second lower face that contacts the actively-cooled plate.

(6) According to another aspect, a method for varying UV intensity emitted by a plurality of UV light emitters is provided. The method is performed using a first UV light emitter support seating the plurality of UV light emitters and a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support. The method includes energizing the plurality of UV light emitters to emit a first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a first gap. The method includes moving the first UV light emitter support away from the second UV light emitter support until the first UV light emitter support is spaced from the second UV light emitter support by a second gap greater than the first gap. The method further includes energizing the plurality of UV light emitters to emit a second UV light intensity greater than the first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a second gap.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a perspective view of a disinfecting system within a lavatory according to examples of the present disclosure.

(2) FIG. 2 shows a schematic diagram of the disinfecting system of FIG. 1 according to examples of the present disclosure.

- (3) FIG. 3 shows one example of an ultraviolet (UV) light-emitting assembly housed in a module according to examples of the present disclosure.
- (4) FIG. 4 shows one example of a UV light-emitting assembly according to examples of the present disclosure.
- (5) FIG. 5 shows an exploded view of the UV light-emitting assembly of FIG. 4.
- (6) FIG. 6 shows another view of the UV light emitter supports and heat sinks of FIG. 4 according to examples of the present disclosure.
- (7) FIG. 7 shows another example of a UV light-emitting assembly according to examples of the present disclosure.
- (8) FIG. 8 shows an exploded view of the UV light-emitting assembly of FIG. 7.
- (9) FIG. 9 shows another example of a UV light-emitting assembly including a moving mechanism according to examples of the present disclosure.
- (10) FIG. 10 shows the UV light-emitting assembly of FIG. 9 with the UV light emitter supports and heat sinks moved closer to one another.
- (11) FIG. 11 shows another example of a UV light-emitting assembly including a moving mechanism according to examples of the present disclosure.
- (12) FIG. 12 shows the UV light-emitting assembly of FIG. 11 with one UV light emitter support and heat sink moved closer to the other.
- (13) FIG. 13 shows another example of a UV light-emitting assembly including a moving mechanism according to examples of the present disclosure.
- (14) FIG. 14 shows a block diagram of an example method for varying UV light intensity emitted by a plurality of UV light emitters according to examples of the present disclosure.
- (15) FIG. 15 depicts an aircraft environment in which UV light-emitting assemblies of the present disclosure are installed.

DETAILED DESCRIPTION

- (16) In view of the considerations discussed above, FIGS. 1 and 2 show one example of a system for disinfecting one or more components using ultraviolet (UV) light-emitting assemblies of the present disclosure. As described in more detail below, in some examples the system utilizes UV light-emitting assemblies incorporating one or more heat sinks that provide heat transfer functionality to enable the assemblies to operate at higher power and provide correspondingly higher UV irradiation. In some examples described below, the assemblies may be mechanically controlled to vary the UV light intensity emitted by the UV light emitters.
- (17) FIG. 1 illustrates a perspective view of a lavatory **102** that includes a system **100** for disinfecting one or more components using ultraviolet (UV) light. The system **100** includes a plurality of UV light-emitting modules **104** containing UV light-emitting assemblies as described further below.
- (18) In the example of FIG. 1, three UV light-emitting modules **104a**, **104b**, and **104c** are shown. The system **100** also includes a power supply module **106** that is electrically connected to each of the UV light-emitting modules **104** and provides power to the UV light-emitting assemblies inside the modules to generate UV light for disinfecting and/or sanitizing components and their surfaces in the lavatory **102**.
- (19) In other examples, the system **100** utilizes fewer or more than three UV light-emitting modules **104** that are electrically connected to the power supply module **106**. In still other examples, the system **100** and/or individually powered UV light-emitting modules **104** can be utilized in a variety of environments, including but not limited to kitchens, galleys, retail establishments, medical facilities, arenas, places of worship, banquet halls, theatres, concert venues, commercial businesses, factories, and other spaces. In some examples, the system **100** and/or individually powered UV light-emitting modules **104** can be utilized in aircraft, spacecraft, and other vehicles, such as buses, trains, marine vessels, and the like.
- (20) In a commercial aircraft, the system **100** can be located within a cabin, galley, crew rest area,

assembly area, cargo area, flight deck, lavatory, and other areas in which individuals, passengers, flight crew, ground crew, and/or maintenance personnel may be located. In the present example of FIG. 1, the lavatory **102** can be located within a vehicle, such as within a cabin of a commercial aircraft. For example, FIG. 15 depicts an aircraft environment in which UV light-emitting modules **104** are installed above passenger seats **1004** in the cabin **1000** of the aircraft.

(21) FIGS. 2 and 3 illustrate one example of a module **104** in which UV light-emitting assemblies of the present disclosure may be enclosed. Module **104** is provided merely as one example, and any suitable housing or enclosure may be utilized with the UV light-emitting assemblies described herein. In other examples, one or more UV light-emitting assemblies may be utilized in a portable device, such as a wand, that is configured to be held by a user. In some examples, such a portable device is also configured to be removably mounted to a support structure, such as a wall.

(22) Returning to the example of FIG. 1, the UV light-emitting modules **104** are positioned to emit the UV light towards one or more components within the lavatory **102** for disinfecting and/or sanitizing the components. In the illustrated example, the one or more components include a sink **112** and a toilet **110**. In this example, the UV light-emitting modules **104** are positioned to emit UV light towards different components. For example, the first UV light-emitting module **104a** is positioned to emit UV light towards the toilet **110** including a flush actuator **114** (e.g., lever, button, etc.) of the toilet **110**. The second UV light-emitting module **104b** is positioned to emit UV light towards the sink **112** and the surrounding region, such as portions of the faucet **116** and countertop **118**. The third UV light-emitting module **104c** is positioned to emit UV light towards the door (not shown) used to enter and exit the lavatory **102**.

(23) In some examples, two or more UV light-emitting modules **104** are positioned to emit UV light towards a common component. In some examples, two or more UV light-emitting modules **104** are physically adjacent and/or mechanically coupled to one another.

(24) The power supply module **106** is electrically connected to the UV light-emitting modules **104** to provide power to the UV light-emitting assemblies therein. In some examples the power supply module **106** includes processing and/or power modulation circuitry within an enclosure or housing. In different examples the power supply module **106** receives electrical energy from a power source, such as power distribution panel or a battery, and distributes the electrical energy among the UV light-emitting modules **104**.

(25) In the example of FIG. 1, the power supply module **106** is mounted within the lavatory **102** and is electrically connected to the UV light-emitting modules **104** via respective power leads **120**, such as one or more electrical wires or power cables. In other examples, one or more of the UV light-emitting modules **104** are integrated with the power supply module **106** in a common housing.

(26) FIG. 2 illustrates a schematic block diagram of the system **100** according to an example of the present disclosure. In this example, the power supply module **106** receives electrical energy from an external power source **202** that is separate and discrete from the power supply module **106**. In some examples the power source **202** is a vehicle electrical system onboard a vehicle or an electrical system of a building or facility. In other examples, the power source **202** is a battery, a generator, or the like.

(27) In the present example the power supply module **106** is electrically connected to the external power source **202** via a power conditioning circuit **204** and power cables **206** and **208**. In different examples the power conditioning circuit **204** includes one or more rectifiers, power factor correction circuits, and/or capacitors for electromagnetic interference filtering. In other examples, the power conditioning circuit **204** is integrated with the power supply module **106** in a common enclosure, such as a housing of the power supply module.

(28) In this example, the power supply module **106** receives electrical energy from the power conditioning circuit **204** and controls distribution of the electrical energy among the UV light-emitting modules **104**. In this example, the power conditioning circuit **204** receives alternating current (AC) electrical energy from the external power source **202** and converts the AC electrical

energy to DC electrical energy. This DC electrical energy is supplied to the power supply module **106**, which converts the DC electrical energy to AC electrical energy and supplies the AC to the UV light-emitting modules **104** to power the generation of UV light as described in more detail below. In some examples, the power supply module **106** also controls one or more operations of the UV light-emitting modules **104**, such as activating and deactivating the modules, and modulating the power output of the modules.

(29) Additionally, and as described in more detail below, some examples of UV light-emitting assemblies of the present disclosure utilize one or more heat sinks that enable the modules to operate at higher power and provide correspondingly higher UV irradiation than prior UV emitters. Additionally, and in some examples described below, one or more moving mechanisms are utilized to vary the UV light intensity emitted by the UV light emitters of an assembly.

(30) With reference now to FIGS. **3-6**, one example of a UV light-emitting assembly **300** according to the present disclosure is illustrated. In different use case examples, the UV light-emitting assembly **300** can be enclosed in module **104** described above and shown in FIG. **3**, or in a variety of other housings, enclosures, or portable devices. In different use cases, the UV light-emitting assembly **300** and the other examples of UV light-emitting assemblies described herein can be utilized in a UV disinfecting system, such as system **100**, and/or in standalone devices.

(31) As shown in FIG. **3**, in this example the UV light-emitting assembly **300** is housed in module **104** that comprises a face plate **312** including a light-transmitting aperture **316** through which UV light from one or more UV light emitters within the enclosure is transmitted. In different examples, the walls of the module **104** can be fabricated from a plastic material or from a conductive material, such as aluminum. In this example, the UV light-emitting assembly **300** utilizes four UV light emitters **320**. In other examples, fewer or more than four UV light emitters may be utilized in UV light-emitting assemblies according to the present disclosure.

(32) The plurality of UV light emitters **320** is configured to emit 222 nm wavelength UV light. In some examples, the UV light emitters **320** can be excimer lamps that utilize a krypton-chlorine (Kr—Cl) gas mixture provided in the lamp bulb. Such excimer lamps emit UV light having a wavelength of 222 nm that can disinfect and sanitize component surfaces via localized anti-viral and antimicrobial effects. Further, 222 nm wavelength UV light can disinfect and sanitize surfaces without skin damaging effects associated with conventional germicidal ultraviolet (UV) exposure. In other examples, the UV light-emitting assembly **300** can utilize other types of UV emitters and UV lamps. Additionally, and as described in more detail below, the UV light emitters **320** are seated in one or more UV light emitter supports within the module **104**.

(33) As noted above, in the example of FIGS. **3-6** the UV light emitters **320** are seated in V-shaped grooves in a first UV light emitter support **322** and a second UV light emitter support **323** that extend parallel to one another. In some examples, the UV light emitter supports **322**, **323** are fabricated from a conductive material, such as aluminum. In this manner and by seating the UV light emitters **320** in the supports, the emitters are electrically coupled to the supports.

(34) With reference to FIGS. **4-6** showing light-emitting assembly **300**, and in one potential advantage of the present disclosure, in this example a first heat sink **346** is affixed to the first UV light emitter support **322** and a second heat sink **380** is affixed to the second UV light emitter support **323**. Accordingly, and as described in more detail below, heat sinks **346** and **380** provide heat transfer functionality that enables the assembly **300** to operate at higher powers and provide correspondingly higher UV irradiation. In some examples, such higher power enabled by heat sinks **346** and **380** in combination with increased gaps between the UV light emitter supports **322**, **323** generates significantly increased UV light intensity and larger irradiance areas as compared to prior configurations. In this manner, fewer UV light assemblies may be utilized to sterilize a given area.

(35) In this example, the first heat sink **346** comprises a first plurality of fins **350** extending from a first base portion **354** of the first heat sink. The first base portion is affixed to a first bottom face **356** of the first UV light emitter support **322**. Similarly, the second heat sink **380** comprises a

second plurality of fins **384** extending from a second base portion **386** of the second heat sink. The second base portion **386** is affixed to a second bottom face **388** of the second UV light emitter support **323**.

(36) In the present example, the UV light emitter supports and heat sinks are separate components that are affixed to one another. In other examples of a heat sink being “affixed” to a UV light emitter support, the heat sink and UV light emitter support are produced from a single material source or material, such as via metalworking or additive manufacturing.

(37) With reference also to FIG. **6**, each fin **350** of the first plurality of fins comprises a first distal end **352** opposite to the first base portion **354**, and these first distal ends are contacting a thermally conductive and electrically insulating plate **370**. Similarly, each fin **384** of the second plurality of fins comprises a second distal end **390** opposite to the second base portion **386**, with these second distal ends also contacting the thermally conductive and electrically insulating plate **370**. In some examples, the thermally conductive and electrically insulating plate **370** is fabricated from a fluoropolymer material, such as polytetrafluoroethylene (PTFE). Accordingly, in these examples the thermal conductivity of the plate **370** further facilitates heat transfer from the UV light emitter supports **322**, **323** via first heat sink **346** and second heat sink **380** to cool the UV light emitters **320**. Additionally, fluoropolymer materials have properties that reflect 222 nm UV light.

Accordingly, this configuration also provides a larger surface area of 222 nm UV light reflective material from which UV light emitted by the UV light emitters **320** is reflected.

(38) Additionally, the gap **377** between first UV light emitter support **322** and second UV light emitter support **323** (and correspondingly between first heat sink **346** and second heat sink **380**) can be widened to maximize the amount of gas mixture in the light emitter bulbs that is excited, and thereby increase the emitted UV light. In one example and with reference to FIG. **4**, the first UV light emitter support **322** and second UV light emitter support **323** (and first heat sink **346** and second heat sink **380**) are positioned with a gap **377** that places both ends of each of the UV light emitters **320** substantially flush with the first outer side **357** and the second outer side **359** of the UV light emitter support **322**, **323**, respectively.

(39) In some examples, the gap **377** between first UV light emitter support **322** and second UV light emitter support **323** can be significantly wider than in prior configurations. In one example, the gap **377** is approximately 17 mm. In this example and where the power supplied to the UV light-emitting assembly **300** is 100 W, the assembly generates approximately 9 mW/cm² over approximately 29.4 cm². By comparison, this example area irradiated by UV light is approximately 29% larger than the area irradiated by the same components configured with a 6 mm gap between first UV light emitter support **322** and second UV light emitter support **323**.

Advantageously, utilizing such an increased gap coupled with the heat dissipation functionality of the present configurations enables these configurations to utilize increased power supplies to provide more effective delivery of UV radiation to wider surface areas. In other examples and in other configurations, gaps between the first UV light emitter support **322** and second UV light emitter support **323** can be greater than 17 mm.

(40) With reference to FIGS. **5** and **6**, the first UV light emitter support **322** comprises a first inner support face **360** that faces a second inner support face **392** of the second UV light emitter support **323**. Similarly, the first heat sink **346** comprises a first inner heat sink face **362** (including inner surfaces of the fins **350**) that faces a second inner heat sink face **394** (including inner surfaces of the fins **384**) of the second heat sink **380**. As shown in FIGS. **4** and **6**, the first inner support face **360** is substantially flush with the first inner heat sink face **362**, and the second inner support face **392** is substantially flush with the second inner heat sink face **394**. Advantageously, and particularly as power to the UV light emitter supports **322**, **323** is increased, this configuration prevents electrical arcing between the first heat sink **346** and second heat sink **380**.

(41) In examples where the UV light emitter supports **322**, **323** are fabricated from a conductive material, such as aluminum, the supports are electrically coupled to a power source via lead wires

365 and **395**, respectively. In some examples the power source is the power supply module **106** of system **100**.

(42) In other examples, the UV light emitter supports **322**, **323** can be fabricated from a fluoropolymer, such as polytetrafluoroethylene (PTFE). In these examples, the UV light emitters **320** are directly coupled to a power source via lead wires connected to terminals at each end of the of the emitters.

(43) With reference now to FIGS. **7-9**, in some examples a UV light-emitting assembly **400** of the present disclosure utilizes a heat sink in the form of an actively-cooled plate **404**. In one example, the actively-cooled plate **404** is fabricated from a thermally-conductive material, such as aluminum, and includes embedded tubing **408** through which liquid coolant is circulated. In different examples, a variety of materials, heat-exchanging technologies, and configurations can be utilized for the actively-cooled plate **404**.

(44) In these examples, thermally conductive and electrically insulating pads are located between the UV light-emitting supports and the actively-cooled plate **404**. As shown in FIGS. **7** and **8**, a first thermally conductive and electrically insulating pad **410** includes a first upper face **414** that contacts and is affixed to the first bottom face **356** of first UV light-emitting support **322**. The first thermally conductive and electrically insulating pad **410** also includes a first lower face **418** that contacts an upper surface **406** of the actively-cooled plate **404**. Similarly, a second thermally conductive and electrically insulating pad **430** includes a second upper face **434** that contacts and is affixed to the second bottom face **388** of second UV light-emitting support **322**, and a second lower face **438** that contacts the upper surface **406** of the actively-cooled plate **404**.

(45) In this example, the actively-cooled plate **404** in combination with the first and second thermally conductive and electrically insulating pads **410**, **430** operates to transfer heat from the first and second UV light-emitting supports **322**, **323**. Additionally, and as described above, the gap between first UV light emitter support **322** and second UV light emitter support **323** can be widened to maximize the amount of gas mixture in the light emitter bulbs that is excited, and thereby increase the emitted UV light. In this respect and as shown in FIG. **7**, the first UV light emitter support **322** and second UV light emitter support **323** are spaced apart to create a gap that places both ends of each of the UV light emitters **320** substantially flush with the first outer side **357** and the second outer side **359** of the UV light emitter support **322**, **323**, respectively.

(46) The aluminum UV light emitter supports **322**, **323** can be electrically coupled to a power source in any suitable manner. In other examples, the UV light emitters **320** are directly coupled to a power source via lead wires connected to terminals at each end of the of the emitters.

(47) In other examples, the UV light emitter supports **322**, **323** can be fabricated from a fluoropolymer, such as polytetrafluoroethylene (PTFE). In these examples, the UV light emitters **320** are directly coupled to a power source via lead wires connected to terminals at each end of the of the emitters.

(48) In some examples, assemblies of the present disclosure also are configured to enable real-time variation of the gap between the first UV light emitter support **322** and the second UV light emitter support **323**, and thereby vary the UV light intensity of UV light emitted from the UV light emitters. With reference now to FIG. **9**, in this example the UV light-emitting assembly **300** of FIGS. **4** and **5** also comprises a moving mechanism in the form of a first actuator **450** and second actuator **460**. In this example moving mechanism and as described in more detail below, the first actuator **450** and second actuator **460** are configured to translate the first heat sink **346** and the second heat sink **380**, respectively, relative to the thermally conductive and electrically insulating plate **370**.

(49) In this example, the first actuator **450** includes a first rod **452** that is coupled to the first light emitter support **322**. The first actuator **450** is controlled to translate the first light emitter support **322** and attached first heat sink **346** in a positive and negative x-axis direction. Similarly, the second actuator **460** includes a second rod **462** that is coupled to the second light emitter support

323. The second actuator **460** is also controlled to translate the second light emitter support **323** and attached second heat sink **380** in a positive and negative x-axis direction.

(50) In this manner and in one example shown in FIGS. **9** and **10**, the first actuator **450** is controlled to translate the first UV light emitter support **322** and first heat sink **346** in a positive x-axis direction, and the second actuator **460** is controlled translate the second UV light emitter support **323** and second heat sink **380** in a negative x-axis direction to narrow the gap **377** between the two UV light emitter supports and heat sinks. As the first and second UV light emitter supports **322**, **323** are moved, they slide underneath the UV light emitters **320** such that the ends of the emitters protrude beyond the first outer side **357** and second outer side **359** of the UV light emitter support **322**, **323**, respectively, as shown in FIG. **10**.

(51) With the gap **377** narrowed between the first and second UV light emitter supports **322**, **323**, the distance between electrical coupling locations on each UV light emitter **320** is also narrowed. In this manner, less of the gas mixture in the light emitter bulbs is excited, and the emitted UV light is reduced as compared to the wider gap **377** of FIG. **9**. The frequency, voltage and/or other characteristics of the power supplied to the first and second UV light emitter supports **322**, **323** also can be adjusted to vary the intensity of emitted UV light. In other examples and in different use cases, the first actuator **450** and/or second actuator **460** can be controlled to widen or narrow the gap between the two UV light emitter supports and heat sinks to thereby vary the UV light intensity of UV light emitted from the UV light emitters as desired.

(52) In some examples, the moving mechanism is configured to translate only the first UV light emitter support **322** or the second UV light emitter support **323**. With reference now to FIGS. **11** and **12**, in this example a single actuator **464** is configured to extend and retract rod **466** to translate the second UV light emitter support **323** toward and away from the first UV light emitter support **322** to vary the gap **377** as desired.

(53) In different examples, the actuators described herein can be any suitable type of motion control component, including but not limited to servo motors, stepper motors, and solenoids. In other examples, any other suitable motion control or motion imparting components may be utilized to translate one or more of the UV light emitter supports, including but not limited to gearing mechanisms, chain drives, and belt drives.

(54) In another example and with reference now to FIG. **13**, in this example the UV light-emitting assembly **400** of FIGS. **7** and **8** also comprises a moving mechanism in the form of first actuator **450** and second actuator **460**. In this example the first actuator **450** is configured to translate the first UV light emitter support **322** and first thermally conductive and electrically insulating pad **410** relative to the actively-cooled plate **404**, and the second actuator **460** is configured to translate the second UV light emitter support **323** and second thermally conductive and electrically insulating pad **430** relative to the actively-cooled plate. As described above with respect to FIGS. **9** and **10**, moving the first and second UV light emitter supports **322**, **323** underneath the UV light emitters **320** changes the distance between electrical coupling locations on each UV light emitter **320**, and correspondingly changes the intensity of emitted UV light.

(55) In different examples of UV light-emitting assemblies of the present disclosure, the assemblies can utilize any suitable combinations of features described herein, including but not limited to heat sink features, component materials, and moving mechanisms.

(56) Turning now to FIG. **14**, a method **500** for varying UV light intensity emitted by a plurality of UV light emitters is illustrated. The method **500** is performed using a first UV light emitter support seating the plurality of UV light emitters and a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support.

(57) At **502**, method **500** includes the step of energizing the plurality of UV light emitters to emit a first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a first gap. At **506**, the method **500** includes the step of moving the first UV light

emitter support away from the second UV light emitter support until the first UV light emitter support is spaced from the second UV light emitter support by a second gap greater than the first gap. At **510**, the method **500** includes energizing the plurality of UV light emitters to emit a second UV light intensity greater than the first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by the second gap. At **514**, the method **500** includes the step of translating the first UV light emitter support in a first direction and translating the second UV light emitter support in a second direction opposite to the first direction. At **518**, the method **500** includes translating only the first UV light emitter support.

(58) Further, the disclosure comprises configurations according to the following clauses.

(59) Clause 1. An ultraviolet (UV) light-emitting assembly for disinfecting one or more components, the assembly comprising: a plurality of UV light emitters; a first UV light emitter support seating the plurality of UV light emitters; a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support; a first heat sink affixed to the first UV light emitter support; a second heat sink affixed to the second UV light emitter support; and a thermally conductive and electrically insulating plate contacting the first heat sink and the second heat sink.

(60) Clause 2. The UV light-emitting assembly of clause 1, wherein the first heat sink comprises a first plurality of fins extending from a first base portion of the first heat sink, and the second heat sink comprises a second plurality of fins extending from a second base portion of the second heat sink.

(61) Clause 3. The UV light-emitting assembly of clause 2, wherein the first base portion is affixed to a first bottom face of the first UV light emitter support, and the second base portion is affixed to a second bottom face of the second UV light emitter support.

(62) Clause 4. The UV light-emitting assembly of clause 3, wherein each fin of the first plurality of fins comprises a first distal end opposite to the first base portion, and the first distal ends of the first plurality of fins are contacting the thermally conductive and electrically insulating plate, and wherein each fin of the second plurality of fins comprises a second distal end opposite to the second base portion, and the second distal ends of the second plurality of fins are contacting the thermally conductive and electrically insulating plate.

(63) Clause 5. The UV light-emitting assembly of any of clauses 1-4, wherein the first UV light emitter support comprises a first inner support face that faces a second inner support face of the second UV light emitter support, the first heat sink comprises a first inner heat sink face that faces a second inner heat sink face of the second heat sink, wherein the first inner support face is substantially flush with the first inner heat sink face, and the second inner support face is substantially flush with the second inner heat sink face.

(64) Clause 6. The UV light-emitting assembly of any of clauses 1-5, wherein the plurality of UV light emitters is configured to emit 222 nm wavelength UV light.

(65) Clause 7. An ultraviolet (UV) light-emitting assembly for disinfecting one or more components, the assembly comprising: a plurality of UV light emitters; a first UV light emitter support seating the plurality of UV light emitters; a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support; a heat sink comprising an actively-cooled plate; a first thermally conductive and electrically insulating pad comprising a first upper face that contacts a first bottom face of the first UV light emitter support and a first lower face that contacts the actively-cooled plate; and a second thermally conductive and electrically insulating pad comprising a second upper face that contacts a second bottom face of the second UV light emitter support and a second lower face that contacts the actively-cooled plate.

(66) Clause 8. The UV light-emitting assembly of clause 7, wherein the plurality of UV light emitters is configured to emit 222 nm wavelength UV light.

(67) Clause 9. An ultraviolet (UV) light-emitting assembly for varying UV light intensity emitted

by a plurality of UV light emitters, the assembly comprising: a plurality of UV light emitters; a first UV light emitter support seating the plurality of UV light emitters; a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced by a gap from the second UV light emitter support; and a moving mechanism configured to vary the gap between the first UV light emitter support and the second UV light emitter support to change the UV light intensity of UV light emitted from the plurality of UV light emitters.

(68) Clause 10. The UV light-emitting assembly of clause 9, wherein the moving mechanism is configured to translate the first UV light emitter support in a first direction and translate the second UV light emitter support in a second direction opposite to the first direction.

(69) Clause 11. The UV light-emitting assembly of clause 9 or 10, wherein the moving mechanism is configured to translate only the first UV light emitter support or the second UV light emitter support.

(70) Clause 12. The UV light-emitting assembly of any of clauses 9-11, further comprising a first heat sink affixed to the first UV light emitter support and a second heat sink affixed to the second UV light emitter support.

(71) Clause 13. The UV light-emitting assembly of clause 12, wherein the first heat sink comprises a first plurality of fins extending from a first base portion, and the second heat sink comprises a second plurality of fins extending from a second base portion.

(72) Clause 14. The UV light-emitting assembly of clause 13, wherein the first base portion is affixed to a first bottom face of the first UV light emitter support, and the second base portion is affixed to a second bottom face of the second UV light emitter support.

(73) Clause 15. The UV light-emitting assembly of clause 13 or 14, wherein each fin of the first plurality of fins comprises a first distal end opposite to the first base portion, each fin of the second plurality of fins comprises a second distal end opposite to the second base portion, and the UV light assembly further comprises a thermally conductive and electrically insulating plate **370** contacting the first plurality of fins at their first distal ends and contacting the second plurality of fins at their second distal ends.

(74) Clause 16. The UV light-emitting assembly of clause 15, wherein the moving mechanism is configured to translate the first heat sink relative to the thermally conductive and electrically insulating plate and to translate the second heat sink relative to the thermally conductive and electrically insulating plate.

(75) Clause 17. The UV light-emitting assembly of any of clauses 12-16, wherein the first UV light emitter support comprises a first inner support face that faces a second inner support face of the second UV light emitter support, the first heat sink comprises a first inner heat sink face that faces a second inner heat sink face of the second heat sink, the first inner support face is substantially flush with the first inner heat sink face, and the second inner support face is substantially flush with the second inner heat sink face.

(76) Clause 18. The UV light-emitting assembly of any of clauses 9-11, further comprising: a heat sink comprising an actively-cooled plate; a first thermally conductive and electrically insulating pad comprising a first lower face that contacts the actively-cooled plate and a first upper face that contacts a first bottom face of the first UV light emitter support; and a second thermally conductive and electrically insulating pad comprising a second lower face that contacts the actively-cooled plate and a second upper face that contacts a second bottom face of the second UV light emitter support.

(77) Clause 19. The UV light-emitting assembly of any of clauses 9-18, wherein the plurality of UV light emitters is configured to emit 222 nm wavelength UV light.

(78) Clause 20. A method for varying UV light intensity emitted by a plurality of UV light emitters, the method performed using a first UV light emitter support seating the plurality of UV light emitters and a second UV light emitter support seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support, the method

comprising: energizing the plurality of UV light emitters to emit a first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a first gap; moving the first UV light emitter support away from the second UV light emitter support until the first UV light emitter support is spaced from the second UV light emitter support by a second gap greater than the first gap; and energizing the plurality of UV light emitters to emit a second UV light intensity greater than the first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a second gap.

(79) Clause 21. The method of clause 20, wherein moving the first UV light emitter support away from the second UV light emitter support comprises translating the first UV light emitter support in a first direction and translating the second UV light emitter support in a second direction opposite to the first direction.

(80) Clause 22. The method of clause 21, wherein moving the first UV light emitter support away from the second UV light emitter support comprises translating only the first UV light emitter support.

(81) The subject disclosure includes all novel and non-obvious combinations and subcombinations of the various features and techniques disclosed herein. The various features and techniques disclosed herein are not necessarily required of all examples of the subject disclosure. Furthermore, the various features and techniques disclosed herein may define patentable subject matter apart from the disclosed examples and may find utility in other implementations not expressly disclosed herein.

Claims

1. An ultraviolet (UV) light-emitting assembly for varying UV light intensity emitted by a plurality of UV light emitters for disinfecting one or more components, the assembly comprising: a plurality of UV light emitters; a first UV light emitter support moveably seating the plurality of UV light emitters; a second UV light emitter support moveably seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced by a gap from the second UV light emitter support; and one or more actuators configured to vary the gap between the first UV light emitter support and the second UV light emitter support to change a UV light intensity of UV light emitted from the plurality of UV light emitters by translating at least one of the first UV light emitter support and the second UV light emitter support to cause relative movement between the at least one of the first UV light emitter support and the second UV light emitter support and the plurality of UV light emitters.

2. The UV light-emitting assembly of claim 1, wherein the one or more actuators are configured to translate the first UV light emitter support in a first direction and translate the second UV light emitter support in a second direction opposite to the first direction.

3. The UV light-emitting assembly of claim 1, wherein the one or more actuators are a single actuator configured to translate only the first UV light emitter support or the second UV light emitter support.

4. The UV light-emitting assembly of claim 1, further comprising a first heat sink affixed to the first UV light emitter support and a second heat sink affixed to the second UV light emitter support.

5. The UV light-emitting assembly of claim 4, wherein the first heat sink comprises a first plurality of fins extending from a first base portion, and the second heat sink comprises a second plurality of fins extending from a second base portion.

6. The UV light-emitting assembly of claim 5, wherein the first base portion is affixed to a first bottom face of the first UV light emitter support, and the second base portion is affixed to a second bottom face of the second UV light emitter support.

7. The UV light-emitting assembly of claim 5, wherein each fin of the first plurality of fins comprises a first distal end opposite to the first base portion, each fin of the second plurality of fins

comprises a second distal end opposite to the second base portion, and the UV light-emitting assembly further comprises a thermally conductive and electrically insulating plate contacting the first plurality of fins at their first distal ends and contacting the second plurality of fins at their second distal ends.

8. The UV light-emitting assembly of claim 7, wherein the one or more actuators are configured to translate the first heat sink relative to the thermally conductive and electrically insulating plate and to translate the second heat sink relative to the thermally conductive and electrically insulating plate.

9. The UV light-emitting assembly of claim 4, wherein the first UV light emitter support comprises a first inner support face that faces a second inner support face of the second UV light emitter support, the first heat sink comprises a first inner heat sink face that faces a second inner heat sink face of the second heat sink, the first inner support face is substantially flush with the first inner heat sink face, and the second inner support face is substantially flush with the second inner heat sink face.

10. The UV light-emitting assembly of claim 1, further comprising: a heat sink comprising an actively-cooled plate; a first thermally conductive and electrically insulating pad comprising a first lower face that contacts the actively-cooled plate and a first upper face that contacts a first bottom face of the first UV light emitter support; and a second thermally conductive and electrically insulating pad comprising a second lower face that contacts the actively-cooled plate and a second upper face that contacts a second bottom face of the second UV light emitter support.

11. The UV light-emitting assembly of claim 1, wherein the plurality of UV light emitters is configured to emit 222 nm wavelength UV light.

12. A method for varying UV light intensity emitted by a plurality of UV light emitters, the method performed using a first UV light emitter support moveably seating the plurality of UV light emitters and a second UV light emitter support moveably seating the plurality of UV light emitters, wherein the first UV light emitter support is spaced from the second UV light emitter support, the method comprising: energizing the plurality of UV light emitters using a first power to emit a first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by a first gap; moving the first UV light emitter support away from the second UV light emitter support by causing one or more actuators to translate at least one of the first UV light emitter support and the second UV light emitter support to cause relative movement between the at least one of the first UV light emitter support and the second UV light emitter support and the plurality of UV light emitters until the first UV light emitter support is spaced from the second UV light emitter support by a second gap greater than the first gap; and energizing the plurality of UV light emitters using the first power to emit a second UV light intensity greater than the first UV light intensity when the first UV light emitter support is spaced from the second UV light emitter support by the second gap.

13. The method of claim 12, wherein moving the first UV light emitter support away from the second UV light emitter support comprises causing the one or more actuators to translate the first UV light emitter support in a first direction and causing the one or more actuators to translate the second UV light emitter support in a second direction opposite to the first direction.

14. The method of claim 13, wherein moving the first UV light emitter support away from the second UV light emitter support comprises causing one or more actuators to translate only the first UV light emitter support.

15. The method of claim 12, wherein a first heat sink is affixed to the first UV light emitter support and a second heat sink affixed to the second UV light emitter support.

16. The method of claim 15, wherein the first heat sink comprises a first plurality of fins extending from a first base portion, and the second heat sink comprises a second plurality of fins extending from a second base portion.

17. The method of claim 16, wherein the first base portion is affixed to a first bottom face of the

- first UV light emitter support, and the second base portion is affixed to a second bottom face of the second UV light emitter support.
18. The method of claim 16, wherein each fin of the first plurality of fins comprises a first distal end opposite to the first base portion, each fin of the second plurality of fins comprises a second distal end opposite to the second base portion, and a thermally conductive and electrically insulating plate contacts the first plurality of fins at their first distal ends and contacts the second plurality of fins at their second distal ends.
19. The method of claim 18, wherein the one or more actuators is configured to translate the first heat sink relative to the thermally conductive and electrically insulating plate and to translate the second heat sink relative to the thermally conductive and electrically insulating plate.
20. The method of claim 15, wherein the first UV light emitter support comprises a first inner support face that faces a second inner support face of the second UV light emitter support, the first heat sink comprises a first inner heat sink face that faces a second inner heat sink face of the second heat sink, the first inner support face is substantially flush with the first inner heat sink face, and the second inner support face is substantially flush with the second inner heat sink face.
21. The method of claim 12, wherein a first bottom face of the first UV light emitter support is contacted by a first upper face of a first thermally conductive and electrically insulating pad that comprises a first lower face that contacts an actively-cooled plate of a heat sink, and a second bottom face of the second UV light emitter support is contacted by a second upper face of a second thermally conductive and electrically insulating pad that comprises a second lower face that contacts the actively-cooled plate of the heat sink.
22. The method of claim 12, wherein the plurality of UV light emitters is configured to emit 222 nm wavelength UV light.
-